

# Generative AI-enhanced Energy-Efficient Multi-UAV Collaborative Mobile Edge Computing

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**Abstract**—Mobile edge computing (MEC) has emerged as a key enabler for low-latency and computation-intensive applications. However, conventional terrestrial MEC networks face challenges in coverage, scalability, and dynamic environments. To address these limitations, this paper explores unmanned aerial vehicle (UAV)-assisted MEC, leveraging the mobility and flexibility of UAVs to enhance edge computing performance. However, challenges such as dynamic mobile user association, multi-UAV collaborative trajectory design, and limited communication and computation resources remain significant hurdles. We then formulate a joint optimization problem for user association, trajectory design, and resource allocation for the multi-UAV collaborative MEC network, aiming to minimize system cost considering computation task demands and UAV resource constraints. The problem, modeled as a Mixed Integer NonLinear Program (MINLP), is reformulated as a decentralized partially observable Markov decision process (Dec-POMDP) to capture complex system dynamics. Afterwards, we propose a multi-agent generative-enhanced deep reinforcement learning method that integrates generative diffusion model (GDM) with multi-agent deep reinforcement learning (MADRL) to effectively handle high-dimensional action spaces and enable collaborative multi-UAV decision-making. Extensive simulations using a real-world mobility dataset demonstrate that the proposed method consistently outperforms three baselines in terms of time-average system cost, task completion delay, task satisfactory ratio, and energy consumption ratio when varying the number of UAVs.

**Index Terms**—Mobile Edge Computing (MEC), Unmanned Aerial Vehicle (UAV), Generative Diffusion Model, Multi-Agent Deep Reinforcement Learning.

## I. INTRODUCTION

Mobile edge computing (MEC) [1] [2] has emerged as a transformative paradigm in wireless communication and computing, enabling low-latency, high-bandwidth, and context-aware services by bringing computation resources closer to end-users. By integrating cloud computing capabilities at the network edge, MEC supports a wide range of applications, from real-time IoT analytics [3] to immersive augmented reality [4], while alleviating the burden on centralized cloud infrastructures. Its ability to leverage multiple access technologies, such as 5G, Wi-Fi, and beyond, ensures flexible and efficient resource utilization, enhancing user experience

and network scalability. However, conventional terrestrial MEC deployments face significant challenges, including limited coverage in remote or dynamic environments and infrastructure constraints. These limitations, coupled with susceptibility to network congestion, impede their ability to meet the stringent requirements of emerging applications [5].

To address the limitations of conventional terrestrial MEC paradigm, unmanned aerial vehicle (UAV)-assisted MEC has emerged as a promising solution, leveraging the mobility and flexibility of UAVs to enhance edge computing capabilities [6] [7]. UAV-assisted MEC offers several distinct advantages, including dynamic deployment in areas with limited or no terrestrial infrastructure, such as rural regions, emergency zones, or temporary event sites [8]. The high mobility of UAVs enables rapid repositioning to optimize network coverage and computation resource allocation, ensuring low-latency and reliable service delivery in dynamic environments. Furthermore, UAVs can operate as aerial base stations or edge servers, facilitating seamless integration with multiple access technologies and supporting computation-intensive applications like real-time video analytics and autonomous vehicle coordination. By providing line-of-sight (LoS) communication links [9], UAV-assisted MEC significantly improves signal quality and reduces latency compared to terrestrial systems. These benefits collectively enhance the scalability, adaptability, and efficiency of edge computing, making UAV-assisted MEC a compelling approach for next-generation wireless networks.

Although promising, UAV-assisted MEC network still faces several significant challenges that need further exploration. First, the dynamic mobile user association presents a critical issue. The continuous mobility of both UAVs and users introduces complexity in assigning users to appropriate UAVs. This environment dynamics requires adaptive strategies to ensure seamless connectivity for MEC tasks execution under fluctuating network conditions, such as varying signal strengths and user densities. Then, the multi-UAV collaborative trajectory design poses a complex challenge. Given the uncontrollable movements of mobile users, UAVs must dynamically adjust their trajectories based on the time-varying spatial distribution of users. This requires sophisticated coordination to achieve

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collaborative objectives, such as minimizing computation tasks completion delay and optimizing energy consumption for UAVs. Furthermore, the resource management is significantly constrained by the limited on-board communication and computation resources of UAVs. Efficient allocation strategies are essential to balance task computation demands, spectrum availability, and energy consumption in the complex and dynamic UAV-assisted MEC networks.

To address the aforementioned challenges, we first formulate a joint optimization problem that integrates dynamic mobile user association, multi-UAV trajectory planning, and communication and computation resources allocation, with the goal of minimizing system cost considering the MEC task demands and UAV energy consumption. Then, we propose a **Multi-Agent Generative-enhanced deep Reinforcement Learning** method MAGenRL for efficient UAV decision-making. Specifically, the generative diffusion model (GDM) can capture complex data distributions, generating diverse and robust decision-making policies that enhance exploration in the complex and dynamic MEC environments. Meanwhile, the multi-agent deep reinforcement learning (MADRL) provides adaptive decision-making capabilities, ensuring efficient multi-UAV collaboration and policy optimization. The main contributions of this paper are summarized as follows.

- We formulate a joint optimization problem for user association, trajectory design, and resource allocation in the multi-UAV collaborative MEC network. The objective is to minimize the system cost considering the computation task demands and UAV energy and resource constraints. As the optimization problem is shown to be a Mixed Integer NonLinear Program (MINLP), which is computationally challenging, we reformulate it as a decentralized partially observable Markov decision process (Dec-POMDP) to capture the complex system dynamics.
- We propose a multi-agent generative-enhanced deep reinforcement learning method (MAGenRL), which integrates the denoising diffusion probabilistic model (DDPM) with multi-agent twin delayed deep deterministic policy gradient (MATD3) framework. By leveraging the generative capabilities of the diffusion model, MAGenRL effectively captures complex probabilistic distributions in high-dimensional action spaces. Furthermore, the multi-agent deep reinforcement learning (MADRL) framework enhances collaborative decision-making in complex and dynamic multi-UAV MEC environments.
- Extensive simulations have been conducted in the real-world user mobility dataset. When varying the number of UAVs, the proposed method consistently outperforms three baselines in terms of the time-average system cost, task completion delay, task satisfactory ratio, and energy consumption ratio.

The rest of this paper is organized as follows. Related work is introduced in Section II. The UAV-assisted MEC system is introduced in Section III. In Section IV, we formulate the

optimization problem as an Dec-POMDP. Section V describes the details of the proposed method MAGenRL. The numerical results based on the real-world mobility dataset is presented in Section VI, followed by concluding remarks and further research issues in Section VII.

## II. RELATED WORK

### A. UAV-assisted Mobile Edge Computing

UAV-assisted MEC networks have gained significant attention as a solution to enhance the flexibility and coverage of edge computing systems, particularly in scenarios where terrestrial infrastructure is limited or impractical. Recent studies have explored the potential of UAVs as mobile edge servers or relays to support computation-intensive applications. For example, the authors in [10] optimized the MEC server selection and UAV trajectories to deliver aerial computation services to terminal devices, and proposed an alternating iterative approach based on the block descent method to solve the formulated problem. Similarly, the work in [11] investigated the data traffic offloading, aiming to minimize the total content acquisition delay by optimizing UAV trajectories and transmission power. In [12], the authors introduced an online algorithm that jointly optimizes caching and offloading decisions for UAV-assisted MEC networks. To minimize the overall service delay, the online problem is decomposed into a series of single-slot problems and iteratively optimizes UAV trajectories and task offloading decisions at each time slot. In the UAV-assisted MEC networks, the user association, UAV trajectory design, and resource allocation are critical factors for enhancing system performance. However, these studies have not fully addressed the joint optimization of these factors, limiting the potential of UAV-assisted MEC systems. To this end, in this study, we formulate a joint optimization problem that integrates dynamic user association, multi-UAV collaborative trajectory design, and multi-dimensional resource management to minimize the system cost considering the MEC task demands and UAV energy consumption.

### B. Deep Reinforcement Learning

Deep reinforcement learning (DRL) has been increasingly applied to optimize complex decision-making in wireless networks and edge computing systems. For example, the work in [13] proposed a double-deep Q-learning (DDQN)-based scheme for UAV trajectory optimization and resource allocation. The authors in [14] introduced a data-driven hybrid soft actor-critic (SAC) approach to obtain the service caching decisions. However, their single-agent frameworks may face scalability challenges in our considered multi-UAV collaborative MEC systems. More recently, the works in [15] [16] explored multi-agent DRL (MADRL) methods to address distributed decision-making for multi-UAV systems. These studies demonstrate the potential of MADRL for coordinating UAV swarms. Nevertheless, these approaches may struggle with the exploration efficiency and modeling uncertainty in the dynamic

and complex multi-UAV collaborative MEC systems. To address these limitations, we propose integrating the generative diffusion model (GDM) with MADRL, which enhances robust policy generation and effective collaboration for joint user association, trajectory design, and resource allocation in the multi-UAV collaborative MEC environments.

### III. SYSTEM MODEL

#### A. System Overview

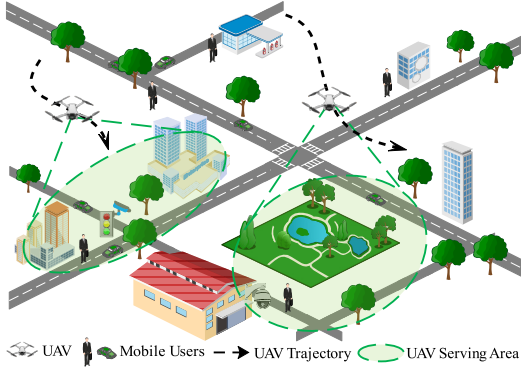


Fig. 1. The system overview of the multi-UAV collaborative MEC scenario.

As shown in Fig. 1, we consider a general multi-UAV collaborative MEC system comprising a set of  $M$  UAVs, denoted as  $\mathcal{M} = \{1, 2, \dots, M\}$ , and a set of  $N$  mobile users, represented as  $\mathcal{N} = \{1, 2, \dots, N\}$ . The total service period for the UAVs is discretized into  $T$  time slots, each of duration  $\tau$ . Without loss of generality, we adopt a 3D Cartesian coordinate system, where the position of UAV  $m$  at time slot  $t$  is given by  $\mathbf{p}_m(t) = [x_m(t), y_m(t), h]^\top$ , with  $h$  denoting the UAV flight height and  $\top$  indicating the transpose operation. Similarly, the position of mobile user  $n$  is presented as  $\mathbf{p}_n(t) = [x_n(t), y_n(t), 0]^\top$ .

Consistent with prior study [17], at each time slot  $t$ , each mobile user  $n$  generates one computing task, denoted as  $\Psi_n(t) = [D_n(t), \Xi_n(t), T_n^{\max}(t)]$ . Here,  $D_n(t)$  represents the task data size (in bits),  $\Xi_n(t)$  denotes the computation intensity (in cycles/bit), and  $T_n^{\max}(t)$  specifies the maximum tolerable delay (in seconds). Meanwhile, at time slot  $t$ , each UAV  $m$  first spends  $\tau_{\text{move},t}^m$  for movement towards a certain direction  $\theta_m(t) \in [0, 2\pi)$  with distance  $d_m(t) \in [0, d_{\max}]$ . Then, UAV  $m$  employs the remaining time  $\tau_{\text{serve},t}^m = \tau - \tau_{\text{move},t}^m$  to provide aerial edge computing services to a subset of mobile users within its sensing range.

#### B. Communication Model

Mobile users can offload their computation tasks to UAVs for processing. To model the air-to-ground uplink propagation channel, we consider both line-of-sight (LoS) and non-line-of-sight (NLoS) components, along with their respective occur-

rence probabilities. Following [9], the average path loss from mobile user  $n$  to UAV  $m$  at time slot  $t$  is expressed as

$$h_{n,m}(t) = 20 \log \left( \frac{4\pi f_c d_{n,m}(t)}{c} \right) + w_{\text{NLoS}} + \frac{w_{\text{LoS}} - w_{\text{NLoS}}}{1 + c_1 \exp \left( -c_2 (\theta_{n,m}(t) - c_1) \right)}, \quad (1)$$

where  $d_{n,m}(t)$  and  $\theta_{n,m}(t)$  represent the Euclidean distance and the evaluation angle between UAV  $m$  and mobile user  $n$ , respectively.  $f_c$  denotes the carrier frequency.  $c$  is the speed of light.  $w_{\text{LoS}}$  and  $w_{\text{NLoS}}$  are the average additional path losses for LoS and NLoS conditions.  $c_1$  and  $c_2$  are environmental-dependent constants (e.g., for rural or urban settings).

We consider orthogonal frequency division multiple access (OFDMA) to ensure that computation task transmissions from scheduled users do not interference with one another. Note that other complicated communication models could be considered but it is not the focus of this study. Let  $a_{n,m}(t)$  represent the bandwidth fraction allocated to user  $n$  at time slot  $t$ . The achievable data transmission rate (in bits per second) from user  $n$  to UAV  $m$  at time slot  $t$  is expressed as

$$R_{n,m}(t) = a_{n,m}(t)W \log_2 \left( 1 + \frac{P_r h_{n,m}(t)}{a_{n,m}(t)W N_0} \right), \quad (2)$$

where  $W$  denotes the total bandwidth available to each UAV,  $P_r$  is the prescribed transmit power, and  $N_0$  is the power spectral density of additive white Gaussian noise.

#### C. Computation Model

The computation tasks can be processed either locally or offloaded to a UAV, which is based on the user association decision. Let  $u_{n,m}(t) \in \{0, 1\}$  denote the user association decision for user  $n$  at time slot  $t$ , where  $u_{n,m}(t) = 1$  indicates that the task is offloaded to UAV  $m$  for processing, and  $u_{n,m}(t) = 0$  indicates local execution.

1) *Local Processing*: When mobile user  $n$  processes task  $\Psi_n(t)$  locally (i.e.,  $u_{n,m}(t) = 0$ ), the local task completion delay is given by

$$T_{\text{loc},n}(t) = \frac{\Xi_n(t) \cdot D_n(t)}{f_{\text{loc},n}}, \quad (3)$$

where  $f_{\text{loc},n}$  is the computation resources of mobile user  $n$ .

2) *Processing at UAVs*: When mobile user  $n$  offloads task  $\Psi_n(t)$  to UAV  $m$  for processing ( $u_{n,m}(t) = 1$ ), the UAV allocates computation resource to execute the task and returns the results to the user. Following [18], the latency of result feedback is typically negligible since the size of the results of significantly smaller than the input data. Then the task completion delay for UAV processing primarily comprises the task transmission delay and UAV task execution delay, which can be calculated by

$$T_{n,m}(t) = \frac{D_n(t)}{R_{n,m}(t)} + \frac{\Xi_n(t) \cdot D_n(t)}{\Lambda_{n,m}(t) f_{\text{uav},m}}, \quad (4)$$

where  $\Lambda_{n,m}(t)$  denotes the computation resource allocation decision by UAV  $m$ , and  $f_{\text{uav},m}$  represents the total computation resource for each UAV.

#### D. Energy Consumption Model

The UAV energy consumption in our considered multi-UAV collaborative MEC system primarily comprises two components, i.e., propulsion energy and task execution energy. The communication-related energy consumption is omitted from this study, as it is considered negligible compared to the propulsion energy [19].

For the UAV propulsion energy consumption, we adopt the energy consumption for rotary-wing UAVs in [20]. Specifically, the propulsion power consumption for UAV  $m$  with velocity  $v_m(t)$  is given by

$$P_{\text{prop},m}^m(t) = P_1 \left( 1 + \frac{3(v_m(t))^2}{v_{\text{tip}}^2} \right) + \frac{1}{2} P_2 (v_m(t))^3 + P_3 \left( \sqrt{1 + \frac{(v_m(t))^4}{4\bar{v}^4}} - \frac{(v_m(t))^2}{2\bar{v}^2} \right)^{\frac{1}{2}}, \quad (5)$$

where  $P_1$ ,  $P_2$ , and  $P_3$  are coefficients related to blade profile power, parasite power, and induced power, respectively.  $v_{\text{tip}}$  denotes the rotor blade tip speed.  $\bar{v}$  is the mean rotor-induced velocity. When UAV  $m$  is hovering (i.e.,  $v_m(t) = 0$ ) for edge computing services, the power consumption simplifies to  $P_{\text{hover}}^m(t) = P_1 + P_3$ . Thus, the total propulsion energy consumption for UAV  $m$  during time slot  $t$  accounting for both flight and hovering phases can be expressed as

$$E_{\text{prop},m}(t) = \tau_{\text{move},t}^m \cdot P_{\text{prop}}^m(t) + \tau_{\text{serve},t}^m \cdot P_{\text{hover}}^m(t). \quad (6)$$

Following [21], the task computation energy consumption for UAV  $m$  can be calculated by

$$E_{\text{comp}}^{n,m}(t) = \varrho \cdot \Xi_n(t) \cdot D_n(t), \quad (7)$$

where  $\varrho$  is the energy consumption per CPU cycle of the UAV. Then the total computation energy consumption for UAV  $m$  at time slot  $t$  is given by

$$E_{\text{comp},m}(t) = \sum_{n \in \mathcal{N}} u_{n,m}(t) E_{\text{comp}}^{n,m}(t). \quad (8)$$

### IV. PROBLEM DEFINITION AND FORMULATION

#### A. Problem Definition

In the multi-UAV collaborative MEC network, the limited on-board energy of UAVs and the latency sensitivity of computation tasks make task completion delay and UAV energy consumption critical performance metrics. Specifically, given the user association decision  $u_{n,m}(t)$ , the task completion delay of task  $\Psi_n(t)$  can be expressed as

$$T_n(t) = \mathbb{1}_{\{u_{n,m}(t)=0\}} T_{\text{loc},n}(t) + \mathbb{1}_{\{u_{n,m}(t)=1\}} T_{n,m}(t), \quad (9)$$

where  $\mathbb{1}_{\{\cdot\}}$  is the indicator function. The energy consumption for UAV  $m$  at time slot  $t$  is given by

$$E_m(t) = E_{\text{prop},m}(t) + E_{\text{comp},m}(t). \quad (10)$$

Then we define a cost function that integrates the task completion delay and UAV energy consumption to evaluate the energy efficiency of the system, which can be denoted by

$$\text{Cost}(t) = c_{\text{comp}} \sum_{n \in \mathcal{N}} T_n(t) + c_{\text{energy}} \sum_{m \in \mathcal{M}} E_m(t), \quad (11)$$

where  $c_{\text{comp}}$  and  $c_{\text{energy}}$  are weight coefficients for task completion delay and energy consumption, respectively, which can be adjusted based on the system preferences for latency and energy consumption.

The objective of the multi-UAV collaborative MEC network is to minimize the time-average system cost in an energy-efficient manner by jointly optimizing the user association  $\mathbf{U} = \{u_{n,m}(t)\}_{n \in \mathcal{N}, m \in \mathcal{M}}$ , communication resource allocation  $\mathbf{A} = \{a_{n,m}(t)\}_{n \in \mathcal{N}, m \in \mathcal{M}}$ , computation resource allocation  $\mathbf{\Lambda} = \{\Lambda_{n,m}(t)\}_{n \in \mathcal{N}, m \in \mathcal{M}}$ , and UAV trajectories  $\mathbf{P} = \{\mathbf{p}_m(t)\}_{m \in \mathcal{M}}$ . Mathematically, the optimization problem can be written as

$$\mathcal{P}1: \min_{\mathbf{U}, \mathbf{A}, \mathbf{\Lambda}, \mathbf{P}} \frac{1}{T} \sum_{t=1}^T \text{Cost}(t) \quad (12)$$

$$\text{s.t. } d_m(t) \in [0, d_{\text{max}}], \theta_m(t) \in [0, 2\pi), \forall m \in \mathcal{M} \quad (13)$$

$$\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| \geq d_{\text{safe}}, \forall i, j \in \mathcal{M}, i \neq j \quad (14)$$

$$T_n(t) \leq T_n^{\text{max}}(t), \forall n \in \mathcal{N} \quad (15)$$

$$u_{n,m}(t) \in \{0, 1\}, \forall n \in \mathcal{N}, m \in \mathcal{M} \quad (16)$$

$$a_{n,m}(t) \in [0, 1], \sum_{n \in \mathcal{N}} a_{n,m}(t) \leq 1, \forall m \in \mathcal{M} \quad (17)$$

$$\Lambda_{n,m}(t) \in [0, 1], \sum_{n \in \mathcal{N}} \Lambda_{n,m}(t) \leq 1, \forall m \in \mathcal{M} \quad (18)$$

Constraint (13) ensures the UAV movement distance and flight angle remain within feasible bounds. Constraint (14) guarantees a safe distance between UAVs. Constraint (15) guarantees the task completion delay not exceed the maximum tolerable delay. Constraint (16) specifies that user association decisions are binary. Constraints (17) and (18) ensures that the communication and computation resource allocations are positive and do not exceed the available resources of UAVs.

The optimization problem  $\mathcal{P}1$  is challenging due to several factors. First, determining optimal decisions requires complete state information, but UAVs are limited to local observations within their sensing range, leading to computational complexity that grows exponentially with the service period and number of UAVs. Second, the presence of continuous variables (e.g., resource allocations  $\{\mathbf{A}, \mathbf{\Lambda}\}$  and trajectory control  $\mathbf{P}$ ) and discrete variables (e.g., user association  $\mathbf{U}$ ) renders  $\mathcal{P}1$  a Mixed-Integer NonLinear Programming (MINLP) problem, which is known to be NP-hard [22]. To address these challenges, we model  $\mathcal{P}1$  as a decentralized partially observable Markov decision process (Dec-POMDP) and propose a MADRL-based approach to solve it.

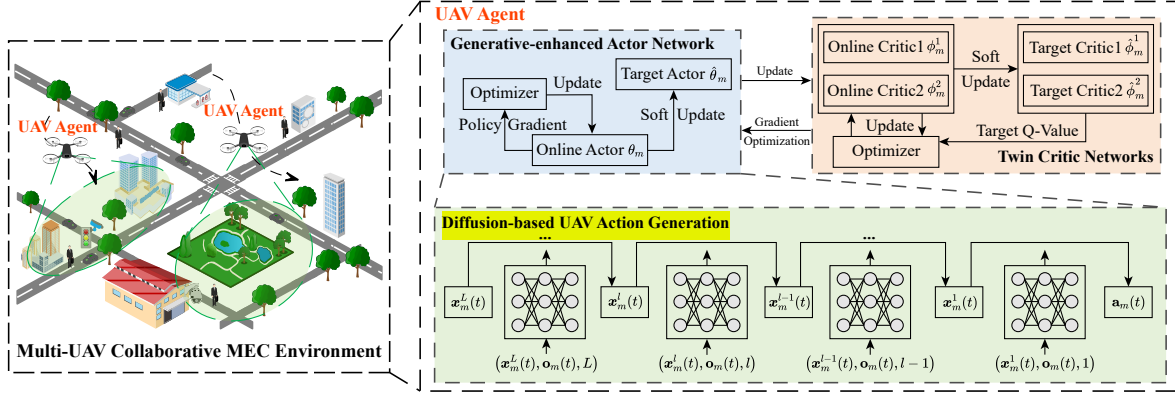


Fig. 2. The framework overview of MAGenRL.

### B. Problem Formulation as Dec-POMDP

To address the optimization problem  $\mathcal{P}1$  effectively, we first model it as a Dec-POMDP for  $M$  UAVs, each acting as an agent. The Dec-POMDP is defined by the six-tuple  $\langle \mathcal{S}, \mathcal{O}, \mathcal{A}, R, Pr, \gamma \rangle$ , where  $\mathcal{S}$  represents the set of the possible environment state of all UAVs,  $\mathcal{O}$  is the local observation space,  $\mathcal{A}$  is the set of available actions for UAVs,  $R$  is the reward function,  $Pr$  is the transition probability, and  $\gamma \in [0, 1)$  denotes the discount factor.

1) *State and Observation Space*: The global state  $\mathbf{s}(t)$  is a vector that concatenates two types of information: (i) the current position  $\mathbf{p}_m(t)$  and remaining energy of each UAV, and (ii) the position  $\mathbf{p}_n(t)$  and computation task demand  $\Psi_n(t)$  of each mobile user.

The local observation  $\mathbf{o}_m(t)$  for UAV  $m$  has the same dimension as the global state. If the mobile users or other UAVs are outside UAV  $m$ 's observable range, the corresponding entries in  $\mathbf{o}_m(t)$  are set to zero, indicating that their information is unavailable.

2) *Action Space*: Each UAV  $m$  selects an action  $\mathbf{a}_m(t)$  based on its local observation  $\mathbf{o}_m(t)$ , which can be defined as

$$\mathbf{a}_m(t) = \{d_m(t), \theta_m(t), \mathbf{U}_m(t), \mathbf{\Lambda}_m(t), \mathbf{A}_m(t)\}, \quad (19)$$

where  $d_m(t)$  and  $\theta_m(t)$  represent the movement distance and radial direction, respectively.  $\mathbf{U}_m(t) = \{u_{i,m}(t)\}_{i \in \mathcal{U}_m(t)}$  denotes the set of dynamically scheduled mobile user set  $\mathcal{U}_m(t)$ .  $\mathbf{A}_m(t) = \{a_{i,m}(t)\}_{i \in \mathcal{U}_m(t)}$  and  $\mathbf{\Lambda}_m(t) = \{\Lambda_{i,m}(t)\}_{i \in \mathcal{U}_m(t)}$  represent the communication and computation resource allocation vectors for the scheduled mobile users.

3) *Reward Function*: To align with the objective of minimizing the cost function in  $\mathcal{P}1$ , the reward function for UAV  $m$  at time slot  $t$  is defined as

$$r_m(t) = - \left( c_{\text{comp}} \sum_{n \in \mathcal{N}} T_n(t) + c_{\text{energy}} E_m(t) \right) + \sum_{n \in \mathcal{N}} \mathbb{1}_{\{T_n(t) > T_n^{\max}(t)\}} \cdot r_p, \quad (20)$$

where  $r_p$  is a constant considered as the penalty to discourage the task completion delay violating constraint (15).

## V. PROPOSED SOLUTION: MAGenRL

### A. Motivation of Adopting GDM and MADRL

The adoption of GDM and MADRL is driven by their ability to optimize decision-making in multi-agent systems. Specifically, the generative capabilities of GDM enable dynamic and adaptive decision-making, while MADRL enhances collaboration in complex, multi-agent environments. To this end, we combine the denoising diffusion probabilistic model (DDPM) [23] with multi-agent twin delayed deep deterministic policy gradient (MATD3) [24] method, and present MAGenRL for efficient multi-UAV decision-making in dynamic and unknown environments. DDPM leverages its iterative denoising process to capture complex data distributions, generating high-quality, diverse samples even with limited interactive data. This capability supports robust and dynamic decision-making in multi-agent settings. Concurrently, MATD3 ensures stable policy updates by mitigating Q-value overestimation through twin delayed critics and clipped double Q-learning, fostering reliable multi-agent collaboration.

### B. Architecture of the Proposed Solution

The architecture of the proposed MAGenRL is shown in Fig. 2. Each UAV is modeled as an independent agent that generates actions based on local observations. Each UAV holds an online actor network, utilizing a diffusion-based model to generate actions, and twin critic networks that evaluate the quality of joint UAV decisions while mitigating Q-value overestimation. To ensure training stability, both the actor and twin critic networks are paired with corresponding target networks. Additionally, the experience replay buffer is employed to store past interactive experiences for training.

1) *Generative-enhanced Actor Network*: In MAGenRL, the actor network  $\pi_{\theta_m}(\cdot)$ , parameterized by  $\theta_m$ , functions as the denoiser within a diffusion model and gradually generates the

**local actions.** The corresponding target actor network is given by  $\pi_{\hat{\theta}_m}(\cdot)$  with network parameter  $\theta_m$ .

In this study, the optimal action  $\mathbf{a}_m^*(t)$  in Eqn. (19) is treated as the optimization solution  $\mathbf{x}_m^0(t)$  within DDPM. During the forward process,  $\mathbf{x}_m^0(t)$  is progressively corrupted with noise over  $L$  steps, indexed by  $l \in \{1, 2, \dots, L\}$ , until it approximates Gaussian noise,  $\mathbf{x}_m^L(t) \sim \mathcal{N}(0, \mathbf{I})$ . The forward process at step  $l$  is expressed as

$$q(\mathbf{x}_m^l(t) | \mathbf{x}_m^{l-1}(t)) = \mathcal{N}(\mathbf{x}_m^l(t); \sqrt{1 - \beta_l} \mathbf{x}_m^{l-1}(t), \beta_l \mathbf{I}), \quad (21)$$

where  $\beta_l \in (0, 1)$  is the variance schedule controlling the noise addition. Since  $\mathcal{P}1$  is an MINLP, it is challenging to obtain the optimal solution  $\mathbf{x}_m^0(t)$ , thus the forward process is not performed in this study and serves to establish the relationship between  $\mathbf{x}_m^l(t)$  and  $\mathbf{x}_m^{l+1}(t)$ .

The actor network serves as a denoiser, tasked with reconstructing  $\mathbf{x}_m^0(t)$  from the noisy input  $\mathbf{x}_m^L(t) \sim \mathcal{N}(0, \mathbf{I})$ , conditioned on the local observation  $\mathbf{o}_m(t)$ . The denoising process follows the reverse diffusion, which is given by

$$p_{\theta_m}(\mathbf{x}_m^{l-1}(t) | \mathbf{x}_m^l(t), \mathbf{o}_m(t)) = \mathcal{N}(\mathbf{x}_m^{l-1}(t); \mu_{\theta_m}(\mathbf{x}_m^l(t), \mathbf{o}_m(t), l), \sigma_l^2 \mathbf{I}). \quad (22)$$

Following [23], we employ an effective parameterization for  $\mu_{\theta_m}$  and  $\sigma_l^2$ , which can be represented as

$$\mu_{\theta_m}(\mathbf{x}_m^l(t), \mathbf{o}_m(t), l) = \frac{1}{\sqrt{\bar{\alpha}_l}} \left( \mathbf{x}_m^l(t) - \frac{\beta_l}{\sqrt{1 - \bar{\alpha}_l}} \epsilon_{\theta_m}(\mathbf{x}_m^l(t), \mathbf{o}_m(t), l) \right), \quad (23)$$

$$\sigma_l^2 = \frac{1 - \bar{\alpha}_{l-1}}{1 - \bar{\alpha}_l} \beta_l, \quad (24)$$

where  $\alpha_l = 1 - \beta_l$ , and  $\bar{\alpha}_l = \prod_{i=1}^l \alpha_i$  denote the cumulative product of  $\alpha_i$ .  $\epsilon_{\theta_m}$  is the noise prediction by the actor network. This process enables the generation of high-quality actions tailored to the local observations of each UAV.

**2) Twin Critic Networks:** In MAGenRL, the twin critic networks  $\{Q_{\phi_m^1}, Q_{\phi_m^2}\}$ , parameterized by  $\{\phi_m^1, \phi_m^2\}$ , takes the system state  $\mathbf{s}(t)$  and the joint UAV actions  $\mathbf{a}(t) = \{\mathbf{a}_m(t)\}_{m \in \mathcal{M}}$  as inputs to compute the Q-value function  $Q_{\phi}(\mathbf{s}(t), \mathbf{a}(t))$ , where  $\phi = \{\phi_m^1, \phi_m^2\}$ . The Q-value function evaluates the expected long-term reward of the state-action pair. To enhance training stability, the corresponding target twin critic networks are denoted by  $\{Q_{\hat{\phi}_m^1}, Q_{\hat{\phi}_m^2}\}$  with network parameter  $\{\hat{\phi}_m^1, \hat{\phi}_m^2\}$ .

**3) Experience Replay Buffer:** In MAGenRL, the experience replay buffer  $\mathcal{D}_r$  is employed to mitigate sample correlation through random sampling during training process. Let  $\mathbf{o}(t) = \{\mathbf{o}_m(t)\}_{m \in \mathcal{M}}$  denote the UAV joint observations,  $\mathbf{r}(t) = \{r_m(t)\}_{m \in \mathcal{M}}$  represent the UAV joint rewards. Then, at each time slot  $t$ , MAGenRL stores the transition tuple  $(\mathbf{s}(t), \mathbf{o}(t), \mathbf{r}(t), \mathbf{s}(t+1), \mathbf{o}(t+1))$  into  $\mathcal{D}_r$ . These experiences are subsequently sampled randomly for the network policy improvements.

**4) Policy Improvement:** The losses of the twin critic networks are computed using the mean-squared temporal difference error over a mini-batch  $B_r$  sampled from  $\mathcal{D}_r$ , which can be expressed as

$$\mathcal{L}(\phi) = \frac{1}{|B_r|} \sum_{t \in B_r} \left( Q_{\phi}(\mathbf{s}(t), \mathbf{a}(t)) - y_m(t) \right)^2, \phi = \phi_m^1, \phi_m^2 \quad (25)$$

where  $y_m(t)$  is the minimum target Q-value outputs of the target twin critic networks.

The policy objective function  $J(\theta_m)$  is used to evaluate the performance of the actor network. Given the identical structure and update mechanism of the twin critic networks, we select the critic network with parameter  $\phi_m^1$  to update the actor network. Therefore,  $J(\theta_m)$  can be denoted by

$$J(\theta_m) = \frac{1}{|B_r|} \sum_{t \in B_r} Q_{\phi_m^1}(\mathbf{s}(t), \mathbf{a}_1(t), \dots, \pi_{\theta_m}(\mathbf{o}_m(t)), \dots, \mathbf{a}_M(t)). \quad (26)$$

To enhance training stability, the parameters of the target networks are soft updated with the corresponding online networks, which can be given by

$$\begin{aligned} \hat{\theta}_m &\leftarrow \omega \theta_m + (1 - \omega) \hat{\theta}_m, \\ \hat{\phi}_m^1 &\leftarrow \omega \phi_m^1 + (1 - \omega) \hat{\phi}_m^1, \\ \hat{\phi}_m^2 &\leftarrow \omega \phi_m^2 + (1 - \omega) \hat{\phi}_m^2, \end{aligned} \quad (27)$$

where  $\omega$  denotes the soft update rate.

### C. Algorithm Description

The overall training procedure of MAGenRL is described in Algorithm 1. MAGenRL employs centralized training and distributed execution (CTDE) mechanism. In the centralized training stage, the actor and twin critic networks, along with the corresponding target networks for all UAVs, and the experience replay buffer are initialized (Lines 1-2). At the beginning of each time slot  $t$ , each UAV generates local actions using its generative-enhanced actor network and interacts with the environment (Lines 5-12). At the end of time slot  $t$ , the resulting transition tuple is stored into the experience replay buffer (Line 13). After collecting enough experiences, UAVs can perform the network parameter updates, with the twin critic networks update at each step and the actor and target networks updated every  $\kappa$  steps to ensuring training stability (Lines 16-23). In the distributed execution stage, each UAV leverages its trained generative-enhanced actor network to generate actions based on its own local observations.

## VI. PERFORMANCE EVALUATION

### A. Simulation Setup

We use Python 3.8.10 and Pytorch 1.7.1 to implement the proposed solution, and all codes are run on Compute Canada [25] with Intel(R) Xeon(R) Silver E5-2650 CPU @ 2.20 GHz, NVIDIA Tesla V100 GPU, and 64 GB memory. To evaluate the performance of MAGenRL, we use the real-world student

**Algorithm 1: MAGenRL Algorithm**


---

```

// Initialization.
1 Initialize the actor network  $\pi_{\theta_m}(\cdot)$  and the twin critic
  networks  $\{Q_{\phi_m^1}, Q_{\phi_m^2}\}$  for each UAV  $m \in \mathcal{M}$ , and
  copy the parameters to the corresponding target
  networks.
2 Initialize the experience replay buffer  $\mathcal{D}_r$ .
3 for episode = 1 to  $E$  do
4   for time slot  $t = 0$  to  $T$  do
      // Diffusion-based UAV Action
      // Generation.
5     for UAV  $m = 1$  to  $M$  do
6       Initialize  $\mathbf{x}_m^L(t) \sim \mathcal{N}(0, \mathbf{I})$ .
7       for denoising step  $l = L$  to 0 do
8         Infer the noise  $\epsilon_{\theta_m}(\mathbf{x}_m^l(t), \mathbf{o}_m(t), l)$  by
          the actor network  $\pi_{\theta_m}$ .
9         Calculate  $\mathbf{x}_m^{l-1}(t)$  with the
          reparameterization technique by
          Eqn. (22).
10        end for
11        Obtain the local action  $\mathbf{a}_m(t) \leftarrow \mathbf{x}_m^0(t)$  for
          each UAV and interact with the
          environment.
12      end for
      // Experience Collection.
13      Collect the state  $\mathbf{s}(t)$ , next state  $\mathbf{s}(t+1)$ , joint
        observations  $\mathbf{o}(t) = \{\mathbf{o}_m(t)\}_{m \in \mathcal{M}}$ , joint next
        observations  $\mathbf{o}(t+1) = \{\mathbf{o}_m(t+1)\}_{m \in \mathcal{M}}$ ,
        joint actions  $\mathbf{a}(t) = \{\mathbf{a}_m(t)\}_{m \in \mathcal{M}}$ , joint
        rewards  $\mathbf{r}(t) = \{r_m(t)\}_{m \in \mathcal{M}}$  of all UAVs.
14      Store experience
         $(\mathbf{s}(t), \mathbf{o}(t), \mathbf{r}(t), \mathbf{s}(t+1), \mathbf{o}(t+1))$  in
        experience replay buffer  $\mathcal{D}_r$ .
15    end for
      // Network Parameter Updating.
16    for UAV  $m = 1$  to  $M$  do
17      Sample a mini-batch  $B_r$  from  $\mathcal{D}_r$ .
18      Update the twin critic networks with Eqn. (25).
19      if time step  $t \bmod \kappa$  then
20        Update the actor network with Eqn. (26).
21        Update target networks by Eqn. (27).
22      end if
23    end for
24  end for

```

---

mobility dataset MDT-NJUST [26], which comprises trajectory data collected on a university campus in Nanjing, China. The selected region in MDT-NJUST dataset spans 1.24 kilometers in length and 1.47 kilometers in width, covering an area of about 1.821 squared kilometers. The dataset includes 55 mobility trajectories generated by students using GPS receivers to track their daily activities.

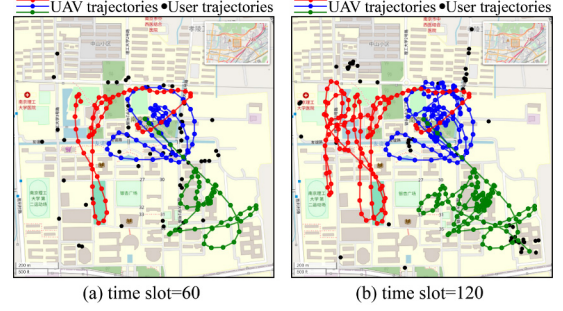


Fig. 3. The UAV trajectories visualization in the MDT-NJUST dataset at time slot  $t = 60$  and  $t = 120$ .

In the simulation, we refer to the parameters of the industrial UAV DJI Mavic 3 Pro [27] to conduct our experiments. The whole serving period for UAV is 30 minutes, which is divided into  $T = 120$  time slots with equal length  $\tau = 15$  s. The maximum on-board energy of the UAV is set to  $E_{\max} = 200$  Wh. Following [17], the total available bandwidth for UAV is set to  $W = 10$  MHz. The computation capacity for each mobile user is randomly selected from  $f_{\text{loc},n} = \{1, 2\}$  GHz, while the computation capacity of UAVs is set to  $f_{\text{uav},m} = 10$  GHz. Based on [20], the UAV energy consumption related parameters  $P_1$ ,  $P_2$ , and  $P_3$  are set to 79.85, 0.018 and 88.63,  $v_{\text{tip}}$  and  $\bar{v}$  are set to 120 m/s and 4.03 m/s, respectively. All UAVs fly at an altitude of  $H_u = 120$  m and the maximum flight speed is set to 15 m/s. To avoid collisions between UAVs, the safe distance is set to  $d_{\text{safe}} = 1$  m. The learning rate for both the actor and twin critic networks is set to  $5 \times 10^{-4}$ .

We perform the performance comparison with three baselines, including TO-MATD3 (Task Offloading-based MATD3) [15], MADDPG [28], and Random. Results are accessed using the time-average system cost calculated by Eqn. (12), task completion delay  $\frac{1}{T \cdot N} \sum_{t=1}^T \sum_{n=1}^N T_n(t)$ , task satisfactory ratio  $\frac{1}{T \cdot N} \sum_{t=1}^T \sum_{n=1}^N \mathbb{1}_{\{T_n(t) \leq T_n^{\max}(t)\}}$ , and energy consumption ratio  $\frac{1}{M \cdot E_{\max}} \sum_{m=1}^M \sum_{t=1}^T E_m(t)$  during the entire aerial edge serving period.

### B. UAV Trajectories Visualization

As shown in Fig. 3, we present the UAV trajectories generated by MAGenRL in the MDT-NJUST dataset with three UAVs deployment. The results show notable collaboration among the UAVs, with each responsible for serving a distinct portion of the selected region. In areas with high mobile user density, UAVs exhibit repetitive back-and-forth movements, driven by their limited sensing range and maximum movement distance per time slot, which renders a single service session insufficient to serve all mobile users in these regions. Notably, the UAVs achieve near-complete coverage of the selected region throughout the service period (i.e., time slot  $t = 120$ ). This can be attributed to the diffusion-based decision-making in MAGenRL, which enhances collaboration and adaptability in dynamic multi-UAV mobile edge computing environments.

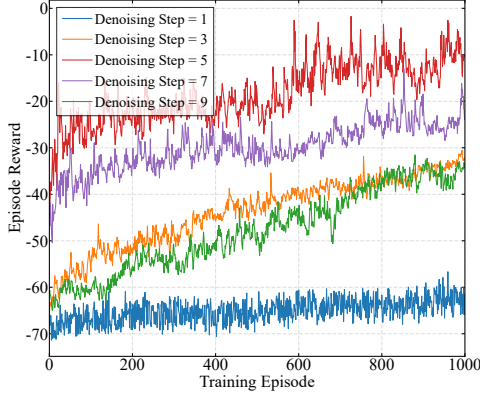


Fig. 4. The impact of the denoising steps in terms of the episode reward.

### C. Simulation Results

1) *Impact of the Denoising Steps:* To investigate the effect of denoising steps on the performance of MAGenRL, we evaluated the number of denoising steps  $L \in \{1, 3, 5, 7, 9\}$  in terms of episode reward for multi-UAV collaborative MEC tasks. As shown in Fig. 4, the episode reward peaks at  $L = 5$ , where MAGenRL effectively balances exploration and exploitation to generate high-quality actions that maximize system performance. When the number of denoising steps is too low (e.g.,  $L = 1$  or 3), the generative-enhanced actor network in MAGenRL fails to adequately refine the initial noisy action distribution, resulting in actions with significant residual noise that impair UAV collaboration and lead to suboptimal decision-making. Conversely, excessive denoising steps (e.g.,  $L = 7$  or 9) over-smooth the action distribution, reducing action diversity and limiting adaptability in dynamic MEC environments, while also increasing computational overhead. To this end, we fix the denoising step  $L = 5$  for MAGenRL in subsequent comparisons with baselines to ensure optimal performance.

2) *Impact of the Number of UAVs:* We demonstrate the impact of varying the number of UAVs,  $M = \{2, 4, 6, 8, 10\}$ , on the time-average system cost, task completion delay, task satisfactory ratio, and energy consumption ratio in the MDT-NJUST dataset, as shown in Fig. 5. We can observe that the proposed MAGenRL consistently outperforms all three baselines. As the number of UAVs increases from two to ten, MAGenRL reduces the system cost from 20.874 to 17.589, while the best baseline TO-MATD3 achieves a reduction from 21.528 to 18.965. Additionally, with ten UAVs deployment, MAGenRL completes nearly all tasks within the maximum tolerable delay, whereas TO-MATD3 achieves only a 90.5% task satisfactory ratio. This performance gain can be attributed to the generative-enhanced actor network in MAGenRL, which enhances multi-UAV collaborative decision-making in dynamic and complex MEC environments through the denoising process for action refinement.

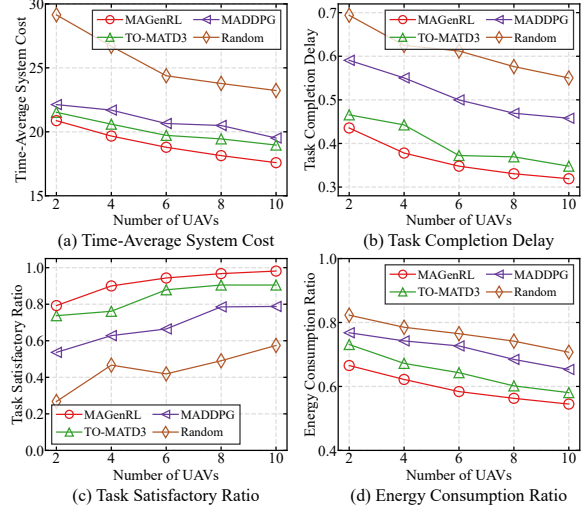


Fig. 5. Performance comparison between the proposed method MAGenRL and three baselines in terms of the time-average system cost, task completion delay, task satisfactory ratio, and energy consumption ratio under different numbers of UAVs in the MDT-NJUST dataset.

### VII. CONCLUSION AND FUTURE WORK

In this paper, we propose MAGenRL, a multi-agent generative-enhanced deep reinforcement learning method for collaborative multi-UAV decision-making in dynamic and complex MEC environments. It consists of a generative-enhanced actor network to generate high-quality actions that maximize system performance through an iterative denoising process. Extensive results using real-world mobility dataset show that MAGenRL outperforms other baselines on the time-average system cost, task completion delay, task satisfactory ratio, and energy consumption ratio during the serving period. In future research, we plan to explore the impact of different user mobility patterns on the performance of multi-UAV collaborative MEC networks.

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